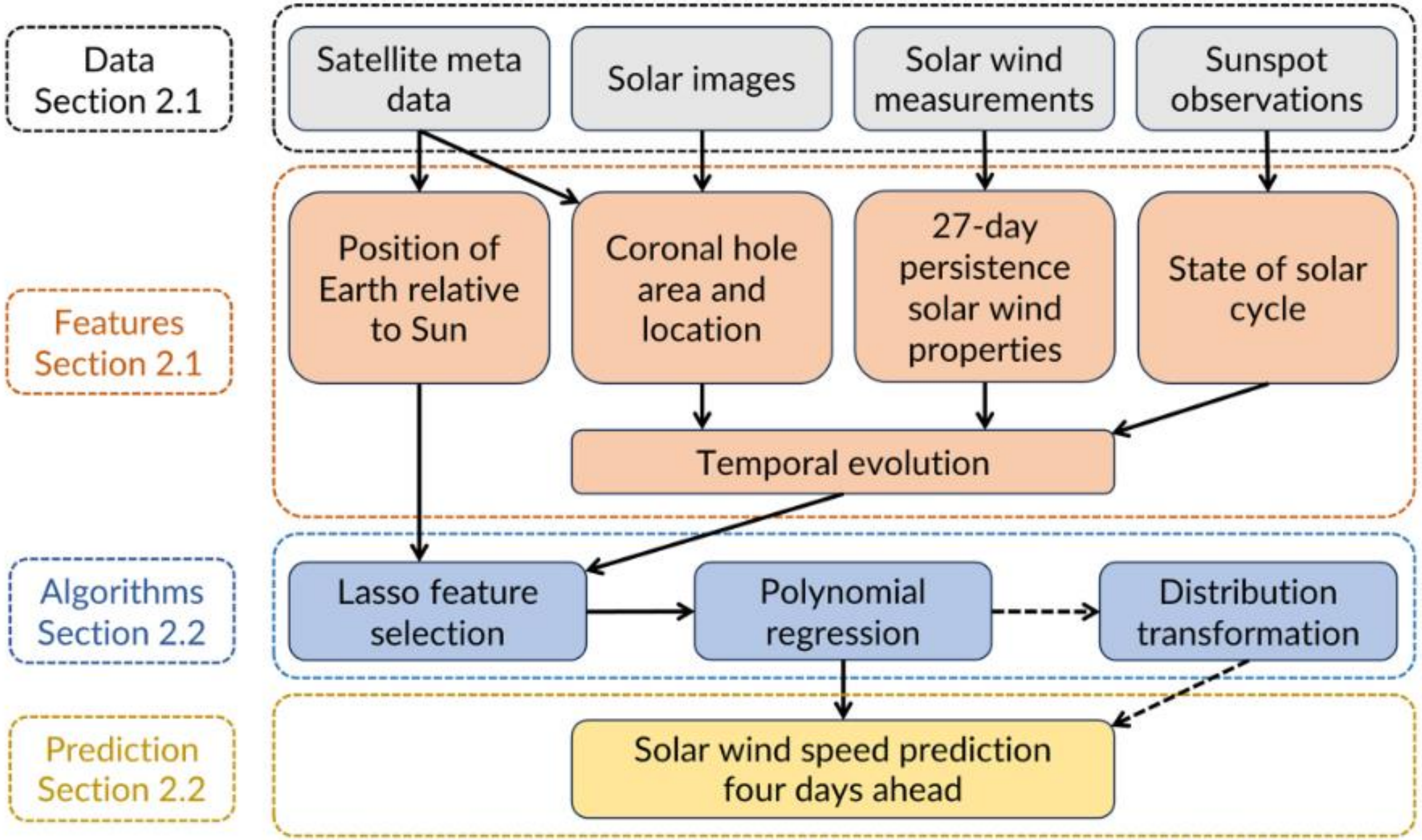
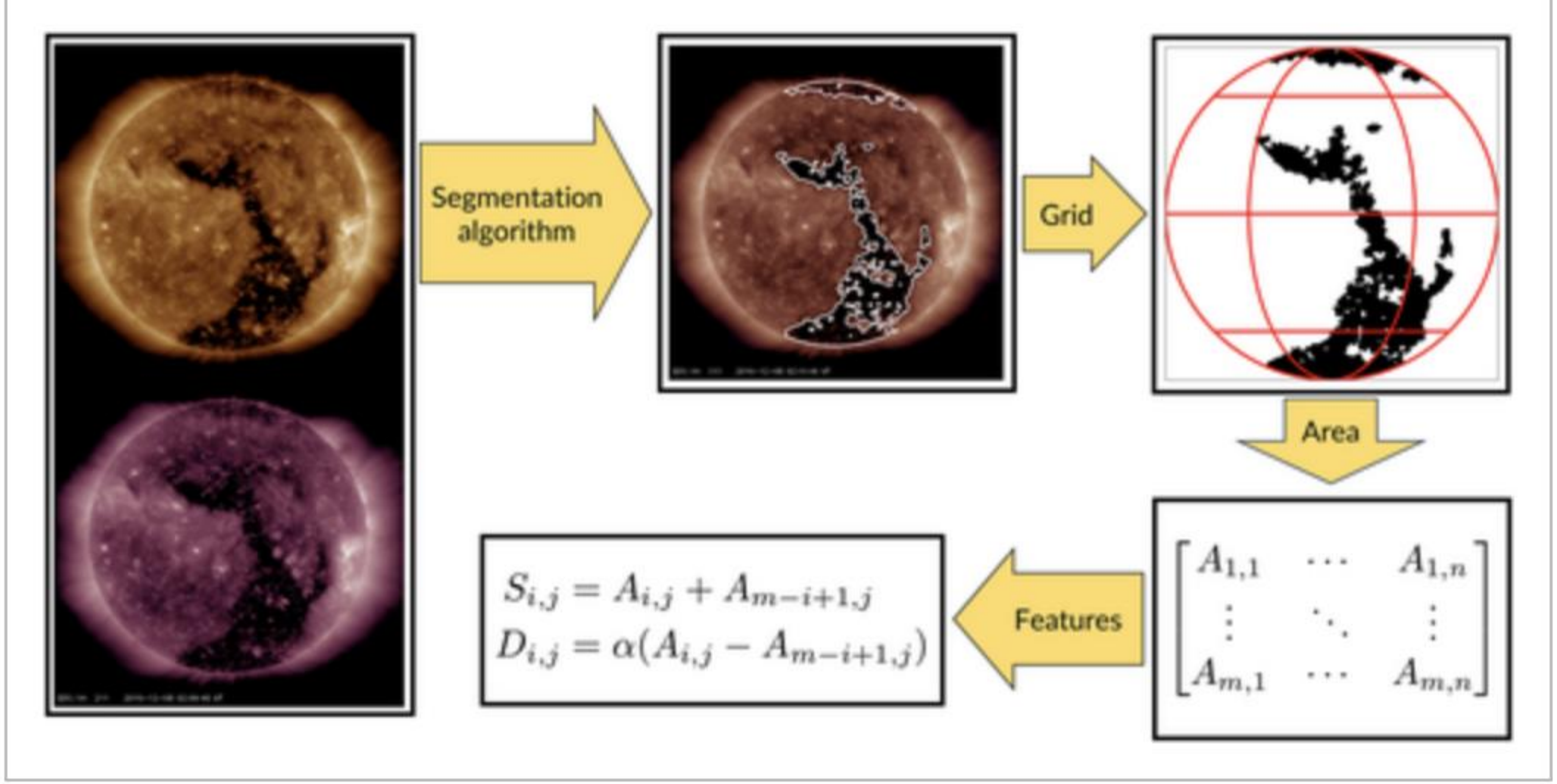
Collin, D., Shprits, Y., Hofmeister, S. J., Bianco, S., & Gallego, G. (2025). Forecasting high-speed solar wind streams from solar images. Space Weather



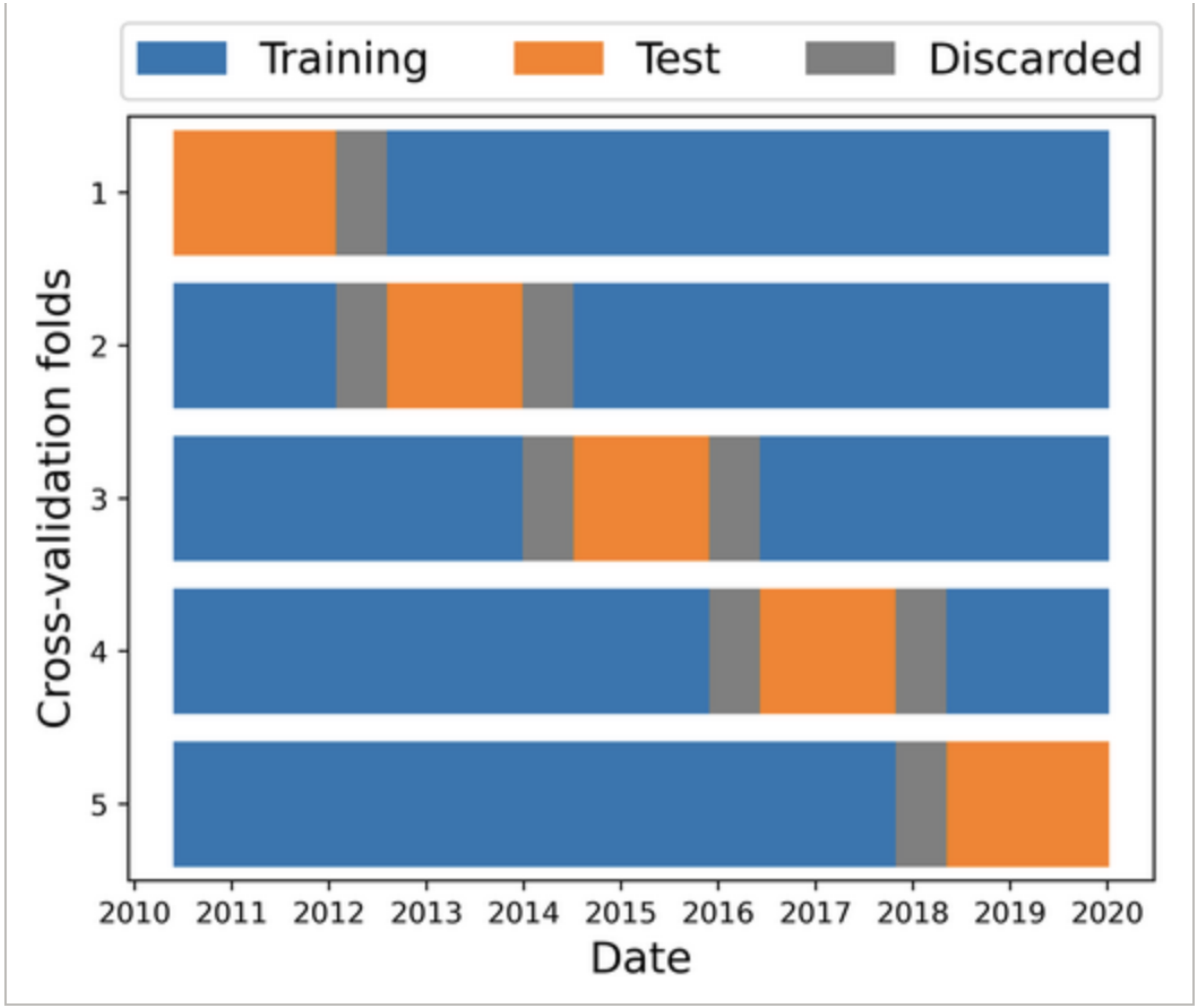


SWS from omniweb

Monthly sunspot number from the NOAA Space Weather Prediction Center

Spacecraft position in heliographic inertial coordinates of satellites at L1 from the OMNI_M data provided by the NASA COHOWeb.

ICMEs affected periods removed. Cadence 1hr to predict 1 hr average SWS.

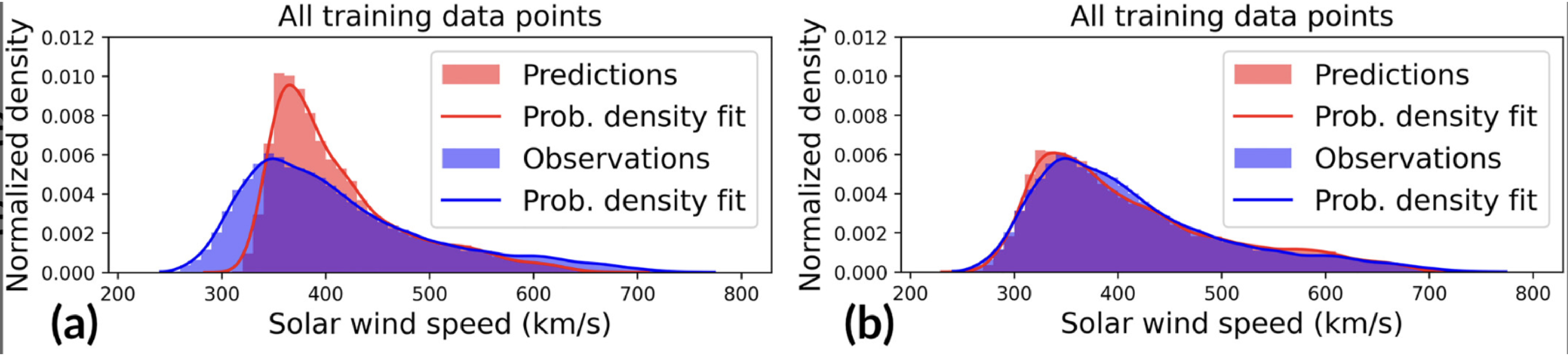


- applying a distribution transformation that maps the distribution of the output of the machine learning model onto the distribution of observations.

- Box-Cox transformation
- Computing lambda maximizing the probability of observing the given data under the assumption that the transformed data follows a normal distribution

$$y^{(\lambda)} \coloneqq F_\lambda(y) = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \text{if } \lambda \neq 0 \\ \log(y), & \text{if } \lambda = 0 \end{cases}, \quad \lambda \in \mathbb{R},$$

$$\mathscr{A} \xrightarrow[\text{transform}]{\text{fit } \lambda_1} F_{\lambda_1}(\mathscr{A}) \xrightarrow{\text{normalize}} \widetilde{F}_{\lambda_1}(\mathscr{A}) \approx \widetilde{F}_{\lambda_2}(\mathscr{B}) \xrightarrow[\text{normalize}]{\text{inverse}} F_{\lambda_2}(\mathscr{B}) \xrightarrow[\text{transform with } \lambda_2]{\text{inverse}} \mathscr{B}$$


All training data points
Normalized density
0.012
0.010
0.008
0.006
0.004
0.002
0.000
200
300
400
500
600
700
800
Solar wind speed (km/s)
Predictions
Prob. density fit
Observations
Prob. density fit
(a)
All training data points
Normalized density
Solar wind speed (km/s)
Predictions
Prob. density fit
Observations
Prob. density fit
(b)

Model	Timeline			HSS peak velocities			HSS events			
	RMSE	MAE	CC	RMSE	MAE	CC	POD	FAR	TS	BS
AV	97.4	76.2	–0.30	–	–	–	–	–	–	–
27	89.1	65.4	0.56	**78.4**	**59.7**	0.54	0.71	0.26	0.57	**0.95**
CH	71.2	54.8	0.67	123.4	102.8	0.52	0.69	**0.11**	0.64	0.78
Ours	**68.1**	**52.6**	**0.70**	87.5	66.8	**0.62**	**0.77**	0.13	**0.69**	0.88
WSA	86.5	64.1	0.38	125.5	100.7	0.10	0.57	0.17	0.51	0.68
ESWF	91.8	69.5	0.43	**104.6**	**81.6**	0.23	**0.68**	0.35	0.50	**1.05**
Ours	**61.8**	**47.4**	**0.55**	121.4	101.4	**0.34**	**0.68**	**0.12**	**0.63**	0.77
EUHFORIA	90.8	68.8	0.36	105.1	87.4	0.28	0.64	0.30	0.50	0.91
Ours	**62.9**	**48.0**	**0.68**	**68.6**	**53.8**	**0.71**	**0.88**	**0.12**	**0.78**	**1.00**
GradBoost	75.6	59.1	0.65	106.6	84.1	**0.60**	0.67	0.16	0.60	0.80
Ours	**71.2**	**55.3**	**0.70**	**95.4**	**73.2**	0.59	**0.74**	**0.14**	**0.66**	**0.86**
SwinTrans	**68.5**	**52.2**	**0.71**	97.3	80.9	**0.71**	0.67	0.25	0.55	0.90
Ours	69.8	54.1	0.70	**86.6**	**68.3**	0.63	**0.79**	**0.17**	**0.68**	**0.96**

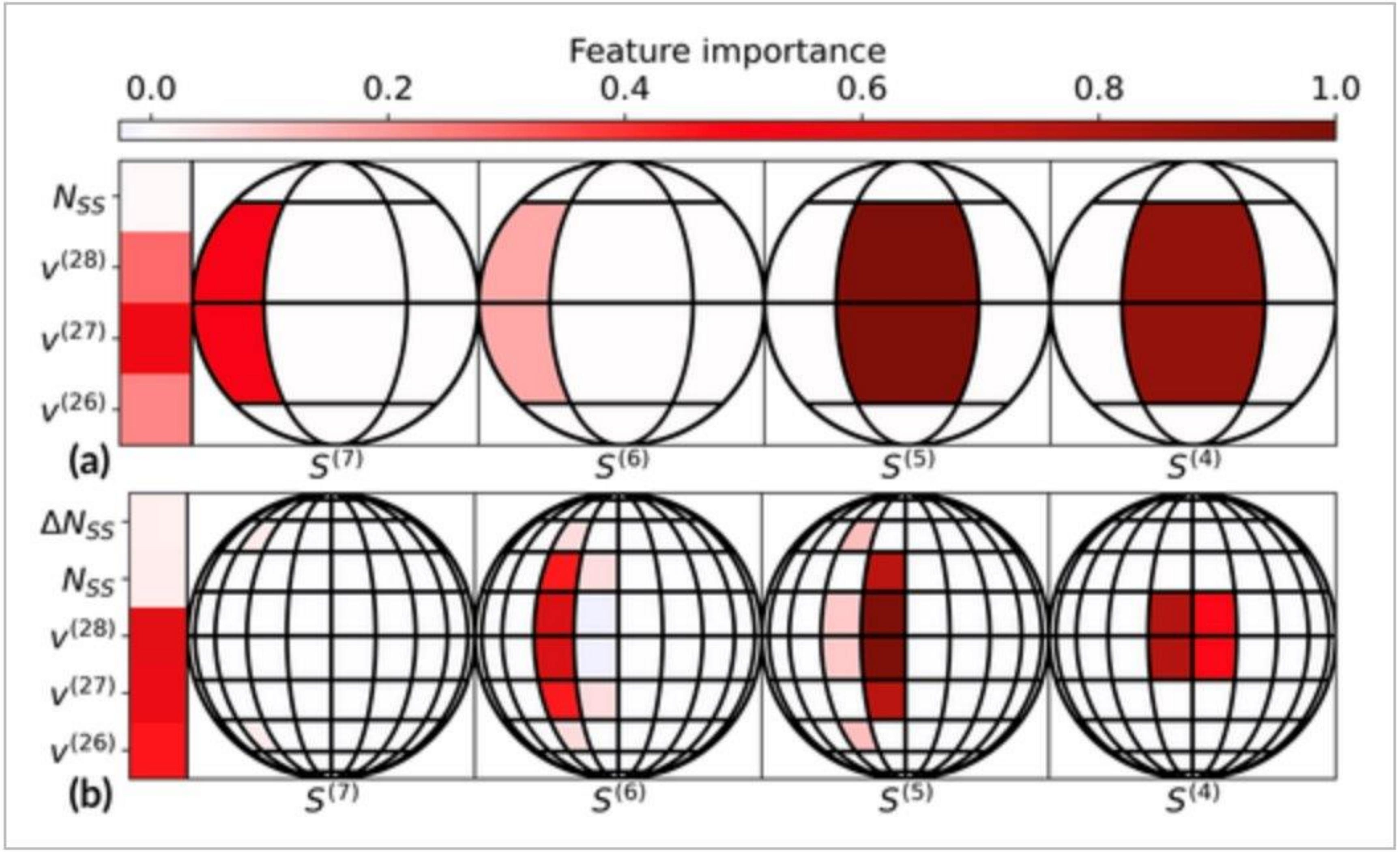
```
Metrics for continuous time series:
                           rmse        mae        me        cc
cv_set eval_data_mode
test   no_cme          68.110930  52.625108  1.978276  0.698873
       trans           75.128345  57.787430  3.212973  0.698729
       with_cme        75.549079  57.981453 -4.910993  0.617423
       trans_with_cme  82.631232  63.426560 -4.230263  0.615576
train  no_cme          63.946301  49.065129  0.000000  0.719512
       trans           69.351576  52.900735  0.142098  0.716694
       with_cme        73.785047  56.077206 -6.991212  0.619740
       trans_with_cme  79.764625  60.691502 -6.384752  0.616220
```

```
Metrics for associated HSS peaks:
                           rmse        mae         me        cc
cv_set eval_data_mode
test   no_cme         113.335817  94.139833 -82.019271  0.581344
       trans           87.468632  66.806237 -51.098932  0.624577
```

	cv_average
S[2,2](-5)	6.152637
S[2,2](-4)	5.669942
speed(-27)	3.447162
S[2,1](-7)	3.255018
speed(-28)	1.774585
speed(-26)	1.499302
S[2,1](-6)	1.092489
sunspot_number	0.054391

```
--- Peak RMSE Importance ---
```

	cv_average
S[2,2](-4)	35.559663
speed(-27)	26.889869
S[2,2](-5)	24.803022
speed(-26)	13.743268
speed(-28)	6.260011
S[2,1](-7)	5.032153
S[2,1](-6)	2.504416
sunspot_number	0.426143

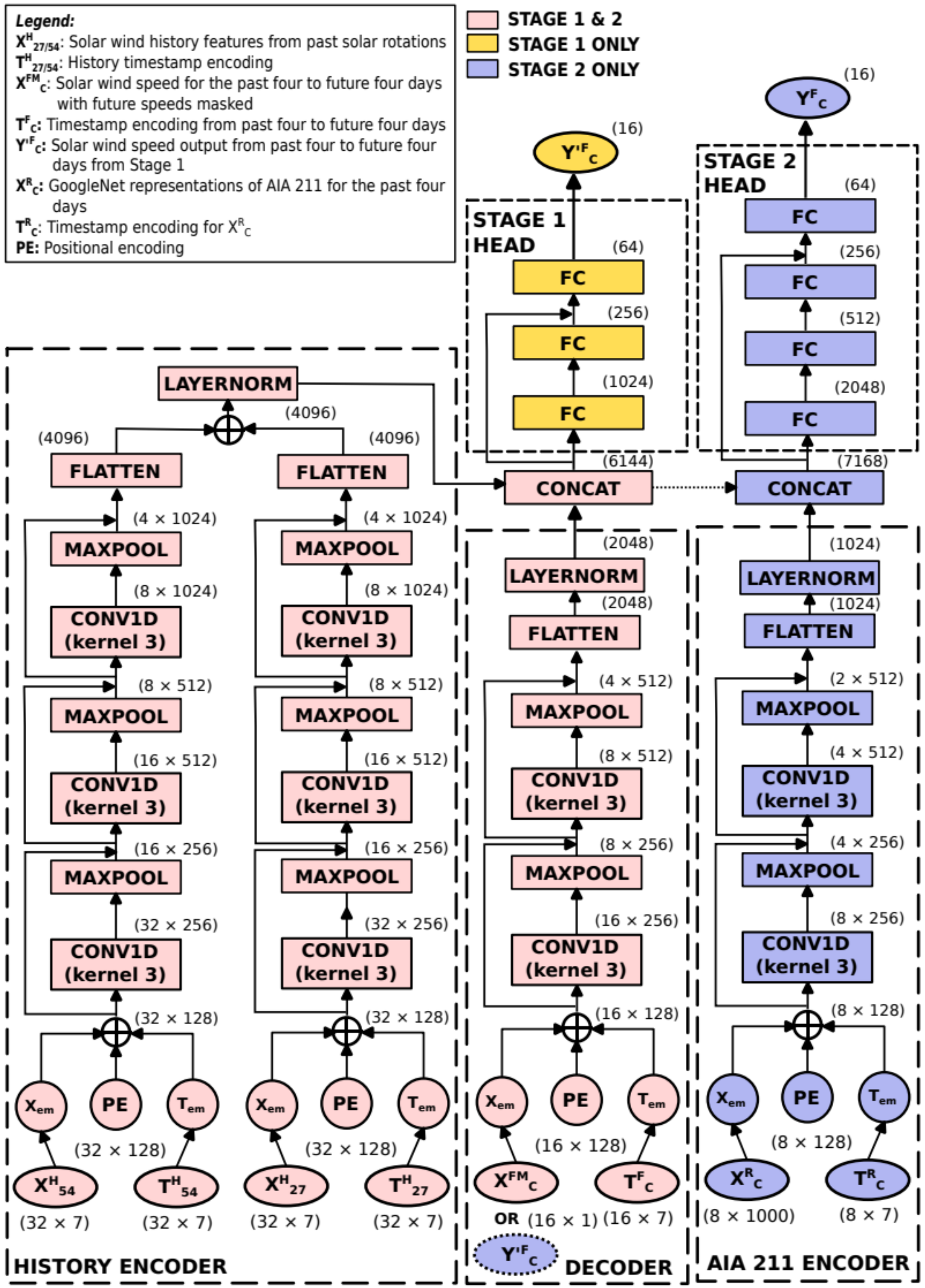

Feature importance
0.0
0.2
0.4
0.6
0.8
1.0
N_{SS}
$v^{(28)}$
$v^{(27)}$
$v^{(26)}$
(a)
$S^{(7)}$
$S^{(6)}$
$S^{(5)}$
$S^{(4)}$
ΔN_{SS}
N_{SS}
$v^{(28)}$
$v^{(27)}$
$v^{(26)}$
(b)
$S^{(7)}$
$S^{(6)}$
$S^{(5)}$
$S^{(4)}$

Multi-modal encoder-decoder neural network for forecasting solar wind speed at L1

Dattaraj B. Dhuri,[1] Shravan M. Hanasoge,[1,2] Harsh Joon,[2] Gopika SM,[2] Dipankar Das,[3] and Bharat Kaul[3]

Using daily and hourly averages sws, 211 AIA from SDOML with 12hr cadence, to predict daily sws

<table>
<tr><th></th><th>Stage 1
(History Encoder Only)</th><th>Stage 2
(History & AIA Encoder)</th></tr>
<tr><td rowspan="4">Training</td><td>01 Aug 1996 – 12 May 2010</td><td>—</td></tr>
<tr><td colspan="2">& 13 May 2010 – 31 Dec 2013</td></tr>
<tr><td colspan="2">& 05 Mar 2017 – 31 Dec 2018</td></tr>
<tr><td>Total samples # : 131272</td><td>Total samples # : 33176</td></tr>
<tr><td rowspan="2">Validation</td><td colspan="2">05 Jan 2014 – 28 Feb 2017</td></tr>
<tr><td>Total samples # : 22788</td><td>Total samples # : 19735</td></tr>
<tr><td rowspan="2">Test</td><td colspan="2">05 Jan 2019 – 01 Aug 2024</td></tr>
<tr><td>Total samples #: 30528</td><td>Total samples #: 24182</td></tr>
</table>


Legend:
$X^H_{27/54}$: Solar wind history features from past solar rotations
$T^H_{27/54}$: History timestamp encoding
X^{FM}_C: Solar wind speed for the past four to future four days with future speeds masked
T^F_C: Timestamp encoding from past four to future four days
Y'^F_C: Solar wind speed output from past four to future four days from Stage 1
X^R_C: GoogleNet representations of AIA 211 for the past four days
T^R_C: Timestamp encoding for X^R_C
PE: Positional encoding
STAGE 1 & 2
STAGE 1 ONLY
STAGE 2 ONLY
HISTORY ENCODER
LAYERNORM
(4096)
FLATTEN
(4 × 1024)
MAXPOOL
(8 × 1024)
CONV1D (kernel 3)
(8 × 512)
(16 × 512)
(16 × 256)
(32 × 256)
(32 × 128)
X_{em}
PE
T_{em}
X^H_{54}
T^H_{54}
X^H_{27}
T^H_{27}
(32 × 7)
DECODER
(2048)
(4 × 512)
(8 × 256)
(16 × 128)
X^{FM}_C
T^F_C
OR
(16 × 1)
(16 × 7)
Y'^F_C
AIA 211 ENCODER
(1024)
(2 × 512)
(4 × 512)
(4 × 256)
(8 × 256)
(8 × 128)
X^R_C
T^R_C
(8 × 1000)
(8 × 7)
CONCAT
(6144)
(7168)
STAGE 1 HEAD
STAGE 2 HEAD
FC
(1024)
(256)
(64)
(2048)
(512)
(16)
Y'^F_C
Y^F_C

Day Relative to Current time	**Stage 1** (History Encoder Only)				**Stage 2** (History & AIA Encoder)			
	Validation		**Test**		**Validation**		**Test**	
	r	RMSE (km/s)	r	RMSE (km/s)	r	RMSE (km/s)	r r	RMSE (km/s)
	Past and Current							
-3	0.99	12.99	0.99	12.33	0.97	26.55	0.96	26.23
-2	0.97	16.87	0.97	15.67	0.95	27.02	0.94	26.64
-1	0.97	15.90	0.97	14.79	0.94	28.58	0.94	30.72
0	0.98	15.26	0.98	13.31	0.94	37.90	0.95	33.78
	Future							
1	0.82	44.50	0.80	39.43	0.78	54.67	0.78	46.88
2	0.62	60.30	0.58	53.21	0.66	58.30	0.61	50.89
3	0.55	63.25	0.51	56.31	0.64	58.17	0.57	52.48
4	0.56	63.25	0.48	57.99	0.63	57.60	0.55	53.25

	This Work: Stage 1		This Work: Stage 2		Upendran et al. (2020)		(Brown et al. 2022)	
Future Day	r	RMSE (km/s)	r	RMSE (km/s)	r	RMSE (km/s)	r r	RMSE (km/s)
4	0.56	63.25	0.63	57.60	0.54	81.21	0.63	72.21
3	0.55	63.25	0.64	58.17	0.55	80.28	—	—
2	0.62	60.30	0.66	58.30	0.52	83.06	—	—
1	0.82	44.50	0.78	54.67	0.54	80.27	—	—